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NOTES

Handling Exceptions in Java -II

Inner try-catch and Multi catch block
Throwing an Exception using throws
Re-Throwing an Exception using throw

Multiple catch Clauses

Using multiple catch blocks allows you to handle different exceptions separately within a single try block.

Example: Multiple catch Clauses

```Java
public class MultipleCatchExample {
    public static void main(String[] args) {
        try {
            int a = args.length;
            int result = 10 / a;
            int[] arr = new int[5];
            arr[10] = 100;
        } catch (ArithmeticException e) {
            System.out.println("Arithmetic Exception: " + e);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Array Index Out Of Bounds: " + e);
        }
    }
}
```

Nested try Statements

Java supports nesting try statements within each other, allowing you to handle exceptions at different levels.

Example of Nested try Statements

```Java
public class NestedTryExample {
    public static void main(String[] args) {
        try {
            int a = args.length;
            int result = 10 / a;
            try {
                int[] arr = new int[5];
                arr[10] = 100;
            } catch (ArrayIndexOutOfBoundsException e) {
                System.out.println("Array Index Error: " + e);
            }
        } catch (ArithmeticException e) {
            System.out.println("Divide by Zero Error: " + e);
```

```
        }
    }
}
```

The 'throw' Keyword

The throw keyword is used to explicitly throw an exception from a method or block of code. This means you can create an exception when a specific condition arises, allowing you to handle errors gracefully.

Real-World Analogy

Think of the throw keyword like a traffic light. When the light turns red, it signals drivers to stop. Similarly, when a certain condition in our code is met (like an invalid input), we can "throw" an exception to indicate that something has gone wrong.

Code Syntax

```Java
throw new ExceptionType("Error Message");
```

Example: Age Validation

```Java
public class AgeValidator {
    static void checkAge(int age) {
        if (age < 18) {
            throw new ArithmeticException("Not eligible to vote.");
        } else {
            System.out.println("Eligible to vote.");
        }
    }
    public static void main(String[] args) {
        checkAge(15);
    }
}
```

In the above case, if someone tries to check their age and they are under 18, the program will throw an exception.

Mechanisms of the 'throw' Keyword

Creating Exceptions: We can create both checked (like IOException) and unchecked exceptions (like IllegalArgumentException) using the throw keyword.

Custom Exceptions: We can define your own exceptions by extending the Exception class.

The 'throws' Keyword

The throws keyword is used in a method's signature to declare that a method can throw exceptions. This informs callers of the method that they need to handle these exceptions.

Real-World Analogy

Imagine you are sending a package that could potentially get lost. By labeling it with "Handle with Care," you are informing others that they should take special precautions. Similarly, using throws informs other parts of your program about potential exceptions.

Code Syntax

```Java
void myMethod() throws IOException {
    // code that might throw IOException
}
```

Code Example

```Java
public class Test {
    java.util.Scanner scan = null;

    void readFile (java.io.File string) throws FileNotFoundException {
         scan = new java.util.Scanner(string);
```

```
    }

    public static void main(String[] args) {
       Test test = new Test()
       test.readFile(new File("Lirary.ser"));
       Scanner scan = new Scanner("Library.ser"); // Asks to handle the
exception
       while (scan.hasNextLine()) {
            System.out.println(scan.nextLine());
        }
    }
}
```

Key Differences Between throw and throws

Feature	**throw**	**throws**
Purpose	Used to actually throw an exception	Used to declare that a method may throw one or more exceptions
Usage	Inside method body	In method signature
Number of Exceptions	Can only throw one exception at a time	Can declare multiple exceptions

Real-Time Usages

User Input Validation: When validating user input, you might use the throw keyword to signal invalid data.

File Operations: When working with files, methods often declare exceptions using throws, allowing callers to handle potential file-related errors.

Network Operations: In network programming, methods may throw exceptions if connections fail or time out.

THANK YOU

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Vishwa Mohan

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vishwa.mohan.singh

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